GOSPEL READING: JOHN 6:22-29

On the next day the people who remained on the other side of the sea saw that there had been only one boat there, and that Jesus had not entered the boat with his disciples, but that his disciples had gone away alone. However, boats from Tiberias came near the place where they ate the bread after the Lord had given thanks. So when the people saw that Jesus was not there, nor his disciples, they themselves got into the boats and went to Capernaum, seeking Jesus. When they found him on the other side of the sea, they said to him, "Rabbi, when did you come here?" Jesus answered them, "Truly, truly, I say to you, you seek me, not because you saw signs, but because you ate your fill of the loaves. Do not labor for the food which perishes, but for the food which endures to eternal life, which the Son of man will give to you; for on him has God the Father set his seal." Then they said to him, "What must we do, to be doing the works of God?" Jesus answered them, "This is the work of God, that you believe in him whom he has sent."

PRAYER

Lord Jesus, you alone can satisfy the deepest longing and hunger in our hearts. May I always hunger for the imperishable bread, that I may be satisfied in you alone as the *True Bread of Heaven*. Nourish and strengthen me that I may serve you with great joy, generosity, and zeal all the days of my life.

St. John Baptist De La Salle

Pray for us

Live Jesus in our hearts

Forever

ATTENDANCE

CONTEXT-FREE GRAMMAR (CFG)

TOPIC LEARNING OUTCOMES

- Recognize the components of CFGs.
- Describe in plain English the languages of CFGs with straightforward descriptions.
- Design CFGs for context-free languages.
- Argue why a language description is or is not the language of a CFG.

CONTEXT-FREE GRAMMARS (CFGS)

- Are syntactic models of languages, similar to regular expressions.
- Through grammar rules, CFGs describe how variables can be replaced by variables and terminals to derive strings in the language of the grammar.

CONTEXT-FREE GRAMMAR (CFG)

A context-free grammar, CFG, is a collection of three things:

1. An alphabet Σ of letters called terminals from which we are going to make strings that will be the words of a language.
2. A set of symbols called nonterminals, one of which is the symbol S, standing for "start here".
3. A finite set of productions or substitution rules of the form $A \rightarrow a$.

CONTEXT-FREE LANGUAGE (CFL)

- The language generated by the CFG.
- The set of all strings of terminals that can be produced from the start symbol S using the productions as substitutions.

EXAMPLES

DETERMINE THE CFL OF THE FOLLOWING CFGS.

1. $S \rightarrow aX$

   $X \rightarrow aX \mid \Lambda$

CFL :

2. $S \rightarrow X$

$X \rightarrow aX \mid aa \mid \Lambda$

CFL :

3. $S \rightarrow XaaX$

$X \rightarrow aX \mid \Lambda$

CFL :

4. $S \rightarrow XY$

$X \rightarrow aX \mid bX \mid a$

$Y \rightarrow Ya \mid Yb \mid a$

CFL :

5. $S \rightarrow SS \mid BS \mid SB \mid USU \mid \Lambda$

$B \rightarrow aa \mid bb$

$U \rightarrow ab \mid ba$

CFL :

CONSTRUCT THE CFG FOR THE FOLLOWING CFLS.

$$\Sigma = \{ a, b \}$$

1. All strings that start with a.

2. All strings with a.

3. All strings ending with a.

4. All strings with double letter.

5. All strings without double letter.

6. All strings with even length.

7. All strings in which the nos. of a is even.

8. All strings without substring ba.

REFERENCES:

- Introduction to Computer Theory by Daniel I. A. Cohen
- Automata Theory by Cecil Jose A. Delfinado